Lab Tasks

Task # 01:

A school wants to manage student records and calculate grades. Design a Java class Student with private data members: rollNo (int), name (String), and marks (double). Provide public setter and getter methods to update and retrieve values, and a method calculateGrade() to determine the student's grade.

Tasks: Create at least three Student objects, update their marks, and display their details along with grades.

Task # 02:

Design a program to help a shopkeeper maintain records of items in the shop. Create a class named Item with two public data members: itemName and itemID, and two private data members: price and stock. Implement appropriate getter and setter methods to access and modify the private data members.

Create at least two objects of the Item class. Allow the user to search for an item they want to purchase and display its price when selected. Use the setter methods to initialize the price and stock of each item. If the user attempts to purchase a quantity greater than the available stock, display an appropriate error message. Finally, calculate and display the total bill for the purchased items while ensuring correct stock updates and pricing.

Task # 03:

A school wants a program to manage student records, including marks and attendance, and determine their results. Design a Java class Student with private data members rollNo (int), name (String), marks (double), and attendance (int). Provide public setter and getter methods to update and retrieve these values, and include a method calculateResult() that prints “Pass” if marks are at least 40 and attendance is at least 75% of total days, otherwise prints “Fail.” Create at least three Student objects dynamically, update their marks and attendance using the setters, and display their details and results, demonstrating proper encapsulation and controlled access to data.

Task # 04:

A fitness center wants a program to track its members and their workout progress. Design a Java class Member with private data members memberID (int), name (String), weight (double), and workoutHours (int). Provide public setter and getter methods to update and retrieve these values, and include a method assessProgress() that prints “Good Progress” if workoutHours ≥ 10 and weight has decreased, otherwise prints “Needs Improvement.” Create at least three Member objects dynamically, update their weight and workout hours using setters, and display their details along with the assessment, demonstrating proper encapsulation and controlled access to data.